

KEEPING STUDENTS FOCUSED ONLINE: AN AI VIRTUAL CLASSROOM FOR ATTENTION MONITORING AND LEARNING GUIDANCE

(IntelliClass AI)



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Declaration

This is to certify that, we declare that our FYP “Keeping Students Focused Online: An AI Virtual Classroom for Attention Monitoring and Learning Guidance” is the result of our own research and development. We have not copied the proposal from anywhere. We declare that in our FYP proposal similarity index is less than 20%.

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1. Introduction

Online and hybrid teaching are now a normal part of higher education in Pakistan and around the world. Platforms such as Zoom, Google Meet and Microsoft Teams allow one instructor to teach forty, fifty or more students who are sitting in their own homes. These platforms solved the problem of *delivering* a lecture at a distance, but they were designed for meetings and conversations, not for teaching. They carry the instructor's voice and slides to the students, and they carry small video tiles back, but they understand nothing about what is happening on the other side of those tiles.

In a physical classroom, an experienced teacher constantly reads the room. A student who is falling asleep, staring out of the window or quietly using a mobile phone is noticed within seconds, and the teacher can respond immediately by asking a question or changing pace. In an online class this natural feedback disappears. The instructor is busy speaking, sharing slides and watching the chat, while the students appear as a grid of thumbnails that is too small and too crowded to read. Students may fall asleep, walk away from the camera, look somewhere else for long periods, or move to a completely different window such as a game, a video or social media, and the instructor usually learns about it only when the results of an assessment are disappointing.

This project proposes an **intelligent virtual classroom** that restores that missing feedback using artificial intelligence. The system is a complete web-based classroom: instructors create a class, upload the lecture slides and teach live over real-time audio and video, while students join with authenticated accounts and roll numbers. In addition to delivering the lecture, and only after each student explicitly grants permission, the system analyses the student's own webcam and shared screen. Computer-vision models estimate face presence, eye state, drowsiness and gaze/head direction, while an OCR and embedding pipeline checks whether the content the student is viewing still matches the slide the instructor is currently teaching.

Two design decisions shape the whole system. First, the facial analysis runs **inside the student's own browser**, so raw video never has to be uploaded or stored; only a few bytes of structured information per second travel to the server. This protects privacy and allows the system to support far more students on modest, free infrastructure. Second, no decision is made from a single video frame. An **alert engine** combines model confidence with how long a condition persists, together with cooldown and grouping rules, so a normal blink or a two-second glance away never produces an alarm, and the instructor receives a short prioritised list of trustworthy alerts instead of a flood of noise.

The system also supports learning after the alert. Lecture audio is transcribed with an open-source speech-to-text model, natural language processing produces summaries, key topics and likely questions, and a retrieval-augmented assistant answers questions using only that lecture's slides and transcript, replying honestly when the material does not cover the question. Instructors receive live dashboards during the class and an analytics report afterwards.

The project therefore brings together computer vision, machine learning, deep learning, natural language processing, generative AI and real-time distributed systems in one working platform, and every component will be judged with measurable results such as precision, recall, F1-score, alert latency, resource usage and performance with up to one hundred concurrent users.

2. Related Work

The proposed system touches four research and product areas: online classroom platforms, automatic student-engagement recognition, drowsiness and gaze estimation, and lecture-content understanding. Relevant work in each area is reviewed below, and the section closes with the research gap this project addresses.

2.1 Online Meeting and Classroom Platforms

Zoom [1], Google Meet [2] and Microsoft Teams [3] dominate online teaching. They provide stable audio and video, screen sharing, chat and recording, and some of them report attendance or basic participation statistics such as time joined or number of chat messages. None of them analyses whether a student is awake, present in front of the camera, or actually looking at the lecture. The responsibility of noticing disengagement is left entirely to the instructor, which is exactly the task that becomes impossible as class size grows.

2.2 Online Examination Proctoring Systems

Remote proctoring tools such as Proctorio [4] observe a candidate’s webcam and screen during an examination and flag suspicious behaviour such as a missing face, a second person, or switching windows. They show that webcam and screen signals can be analysed automatically, but they differ from this project in purpose and in ethics. They are built for short, high-stakes examinations rather than everyday teaching, they record and store complete video of the candidate, and they produce accusations rather than teaching support. Coghlan et al. (2021) [5] warn that such opaque, heavily recorded surveillance damages trust between students and institutions. This project takes the opposite position: the instructor receives supportive alerts, no video is stored, and no student is ever labelled dishonest.

2.3 Automatic Engagement Recognition Research

Academic work on automatic engagement recognition is well developed. Whitehill et al. (2014) [6] showed that student engagement can be recognised from facial expressions with accuracy comparable to human observers, establishing that the task is feasible. Gupta et al. (2016) [7] published DAiSEE, a large dataset of students watching online lectures in unconstrained “in the wild” conditions, labelled for boredom, confusion, engagement and frustration. Kaur et al. (2018) [8] went further and localised engagement intensity within unconstrained videos. A review by Dewan et al. (2019) [9] surveys this field and reaches two conclusions that directly motivate this project: most systems are evaluated **offline on recorded video** rather than inside a live classroom, and very few are integrated into a working teaching platform that an instructor can actually use during a lecture.

2.4 Drowsiness Detection

Drowsiness detection has been studied most intensively for drivers. Soukupová and Čech (2016) [10] introduced the Eye Aspect Ratio (EAR), a simple ratio computed from six eye landmarks that indicates whether the eye is open or closed and can run in real time on ordinary hardware. Combined with PERCLOS (the proportion of time the eyes are closed) and blink-rate analysis, EAR is the standard baseline against which learned models are compared, and public datasets such as the MRL Eye Dataset, Closed Eyes in the Wild and YawDD support training. This project adopts EAR as a baseline but applies it to the online learner rather than the driver, where camera angle, lighting and observation time differ considerably.

2.5 Gaze Estimation and Facial Landmark Tracking

Zhang et al. (2015) [11] introduced MPIIGaze and showed that appearance-based gaze estimation works under real-world lighting and head-pose variation, although accuracy degrades with extreme angles and poor illumination. MediaPipe (Lugaresi et al., 2019) [12] and its Face Mesh model (Kartynnik et al., 2019) [13] made real-time tracking of 468 facial landmarks possible on ordinary laptops, mobile devices and, importantly for this project, inside a web browser. This body of work makes on-device attention estimation practical, and also makes clear that an honest system must include an **uncertain** state instead of forcing a confident answer when conditions are poor.

2.6 Lecture-Content and Document Understanding

Reliable open-source components now exist for understanding what is displayed on a screen. PP-OCR (Du et al., 2020) [14] provides lightweight text recognition, CLIP (Radford et al., 2021) [15] produces joint image–text embeddings, and Sentence-BERT (Reimers & Gurevych, 2019) [16] gives efficient semantic similarity between texts, while Whisper (Radford et al., 2023) [17] offers robust multilingual speech-to-text and retrieval-augmented generation (Lewis et al., 2020) [18] lets a language model answer from an authorised set of documents. These components are mature individually, but the literature does not report a system that uses them to compare a student’s live screen with the slide an instructor is currently teaching.

2.7 Classroom Automation Systems

A related family of systems automates the administrative work of a **physical** classroom; Chen et al. (2020) [19] survey this rapid growth of artificial intelligence in education and note that most deployed systems target administration rather than moment-to-moment learning behaviour. Such projects typically install a classroom camera and a mobile or desktop application, and automate attendance through face recognition, quiz generation and hosting, record keeping and the preparation of notes, often adding periodic engagement reports computed from images captured every few minutes. These systems are valuable, but their setting and objective are fundamentally different from this project. They assume that students and teacher share one room and one camera, so they cannot observe an individual student’s eye state continuously, and they cannot see what is displayed on each student’s personal screen. Their reporting is periodic and administrative, whereas the present project targets second-by-second behaviour during a live remote lecture and reacts while the lecture is still in progress.

2.8 Research Gap

Taken together, the reviewed work leaves a clear gap:

- Meeting platforms deliver online classes but contribute no understanding of student behaviour, while proctoring systems analyse the same signals only for examinations, with heavy recording.
- Engagement research is accurate but offline and dataset-oriented, and drowsiness and gaze methods are mature yet applied mostly to drivers rather than learners.
- Classroom automation systems target physical rooms and administrative tasks, not individual remote learners.
- No reviewed system compares a student’s live screen with the instructor’s current slide, or combines these signals through confidence- and persistence-based alerting.

Table 1 summarises this comparison. Every source cited in this section is listed in Section 10, where the paper or product link is given so that the reader can open the original source directly.

Capability	Meeting platforms	Proctoring tools	Engagement research	Classroom automation	Proposed system
Delivers the live online class	Yes	No	No	No	Yes
Per-student webcam analysis	No	Yes	Yes	No (room camera)	Yes
Drowsiness (eye closure) detection	No	Partial	Partial	No	Yes
Student screen vs. current slide	No	Partial	No	No	Yes
Reacts during the lecture	No	Yes	No (offline)	Periodic	Yes
Confidence + persistence alerting	No	No	No	No	Yes
On-device analysis, no video stored	N/A	No	N/A	No	Yes

Table 1: Comparison of existing solutions with the proposed system

3. Problem Description

In a large online class the instructor must deliver the lecture and, at the same time, judge whether dozens of remote students are actually following it, which is practically impossible when every student is a small video tile. Students may become drowsy, leave the camera, look away for long periods, or move to unrelated content on their own screens, and because the instructor is speaking and presenting, these problems are usually discovered only when assessment results are poor. Existing tools solve only parts of the problem: meeting platforms provide communication but no understanding of attention; proctoring software watches students but is examination-oriented, records complete video and creates serious privacy concerns; engagement-detection research is accurate but is evaluated offline on recorded datasets; and classroom automation systems are built for physical rooms and administrative tasks such as attendance and quizzes. A naive automated solution would also create new problems, because single-frame decisions generate constant false alarms, streaming every student’s video to a server neither scales nor respects privacy, and poor lighting or camera angles make predictions unreliable. What is needed, therefore, is a real-time and privacy-aware virtual classroom that converts permitted camera, screen and audio signals into a small number of trustworthy, prioritised alerts while the lecture is still running, that continues to work when some signals are missing or uncertain, and that supports learning rather than accusing students.

4. Aims and Objectives

Aim

- To build an easy-to-use online classroom that quietly notices when a student has become sleepy, distracted or has moved away from the lecture, informs the teacher only when it really matters, and turns each lecture into notes and answers that help students revise, all without recording or storing any video.

Objectives

1. **Build the online classroom.** Create a website where a teacher can start a class and students can join with their roll number, hear and see the teacher, and follow the slides being taught.
2. **Notice sleepy students.** Use the student's own camera to see when the eyes stay closed for too long, and report drowsiness only when it continues for several seconds. Target: correct in at least 85 out of 100 test cases.
3. **Notice when attention goes away.** Detect when a student looks away from the lecture for a long time or leaves the camera, and honestly report "not sure" when the light or camera angle is too poor to decide.
4. **Check whether the student is on the right screen.** With the student's permission, compare what is open on the student's screen with the slide the teacher is currently teaching, and report a mismatch only if it lasts longer than a short grace period.
5. **Send few alerts, but useful ones.** Never raise an alarm for a blink or a two-second glance, and group repeated problems so the teacher sees a short list instead of a flood of messages. Target: an alert reaches the teacher within 3 seconds.
6. **Let the teacher call one student privately.** When the teacher taps an alert, play a gentle sound and show a short message on that one student's device only, so the rest of the class is not disturbed and the student is not embarrassed in front of others.
7. **Give the teacher a live screen and a report after class.** Show who needs attention right now during the lecture, and prepare an automatic summary report of the class once it ends.
8. **Report with signals, not with pictures.** Send the teacher only short status words for each student, such as *sleepy*, *camera off*, *no face in front of the camera*, *looking away*, *wrong screen* or *connection problem*, instead of any video, photograph or screenshot, so the teacher immediately sees who needs attention while no image of any student ever leaves that student's own device.
9. **Give every student a private summary of their own.** Let each student see only their own attention and participation summary, so the system helps them improve instead of only reporting them.
10. **Turn the lecture into notes.** Convert the teacher's voice into text, prepare a short summary with the key points and likely questions, and let students ask questions that are answered only from that same lecture.
11. **Protect student privacy.** Do the face checking on the student's own laptop, store no video or screen recording, tell students clearly what is collected, and keep a record of who viewed their information.
12. **Start simple, then prove the improvement.** Build a basic working version of each feature first, then improve it and show with test results and numbers that the improved version is really better.
13. **Test it properly.** Test the system with our own recorded samples, with up to 100 students joining at the same time, and with real teachers and students who try it and give feedback.

5. Domain of the Project

This is a **development project** in the domain of **Artificial Intelligence–based web application development for educational technology (EdTech)**. It combines Computer Vision, Machine Learning and Deep Learning, Natural Language Processing and Generative AI with real-time communication (WebRTC) and a containerised, service-oriented cloud architecture.

6. Project Scope

The project will deliver a working web-based virtual classroom for live online lectures, with on-device vision analysis (face presence, eye state and drowsiness, gaze and head direction), screen-to-slide compliance checking, a confidence- and persistence-based alert engine, a private one-to-one nudge from the instructor to a single student, an instructor dashboard with live alerts and post-class analytics, lecture transcription with summaries and a grounded assistant, and a containerised deployment evaluated with up to one hundred concurrent participants.

The following aspects are outside the scope of this project:

- Examination proctoring or any automatic decision about cheating or misconduct; the system raises alerts only, and every decision remains with the instructor.
- Administrative classroom automation such as quiz generation, examination hosting, timetable or fee management; attendance is only a by-product of presence detection, not a module of this project.
- Emotion recognition and face-recognition-based identity verification; students are identified by their authenticated account and roll number.
- Recording or storing raw video or continuous screen recordings; only event metadata and the instructor’s lecture material are stored.
- Operating-system level screen locking or bypassing browser permission indicators, which a web application cannot and should not do.
- Full feature parity with commercial meeting applications (breakout rooms, whiteboards, cloud recording) and native mobile applications.

Assumptions: students use a laptop or desktop with a webcam and a modern Chromium-based browser; they explicitly consent to camera and screen sharing; the instructor uploads lecture slides (PDF/PPT) before the class; the internet connection is stable enough for a video call; and lectures are delivered mainly in English, optionally mixed with Urdu, which the speech-to-text model supports.

7. Methodology

The project follows an **Agile incremental model** with two-week sprints. Each sprint produces a working increment that is demonstrated to the supervisor, so feedback arrives early. For every AI component a simple baseline is built first, measured, and only then replaced or extended by a learned model, so that every claimed improvement is supported by evidence.

7.1 Classroom Platform and User Management

- Instructors and students register and log in; JWT authentication and role-based access control protect every API endpoint.

- Instructors create classes, enrol students by roll number and upload the lecture slides that later serve as the compliance reference.
- Before joining, each student sees a clear notice explaining which signals are analysed, and grants camera, microphone and screen permissions explicitly.
- Audio and video are carried over WebRTC through a Selective Forwarding Unit, which scales far better than peer-to-peer connections, while a secure WebSocket channel backed by Redis carries events and alerts.

7.2 On-Device Vision Pipeline for Drowsiness and Attention

The first objective is handled entirely inside the student’s browser using MediaPipe Face Mesh (Kartynnik et al., 2019) [13]. Camera frames are sampled at five to ten frames per second and 468 facial landmarks are extracted. From these landmarks the system computes the Eye Aspect Ratio (Soukupová & Čech, 2016) [10], blink rate and PERCLOS for drowsiness, and head pose together with gaze direction for attention. Only the resulting numerical features leave the device. A temporal classifier on the server then decides the student’s state over a sliding window, so that short blinks and quick glances are not confused with genuine drowsiness or distraction, and an *uncertain* state is produced when lighting or camera angle make the estimate unreliable. The rule-based EAR baseline will be compared against a learned classifier trained on public datasets such as the MRL Eye Dataset, YawDD and DAiSEE (Gupta et al., 2016) [7], together with a small test set recorded by the team with consent.

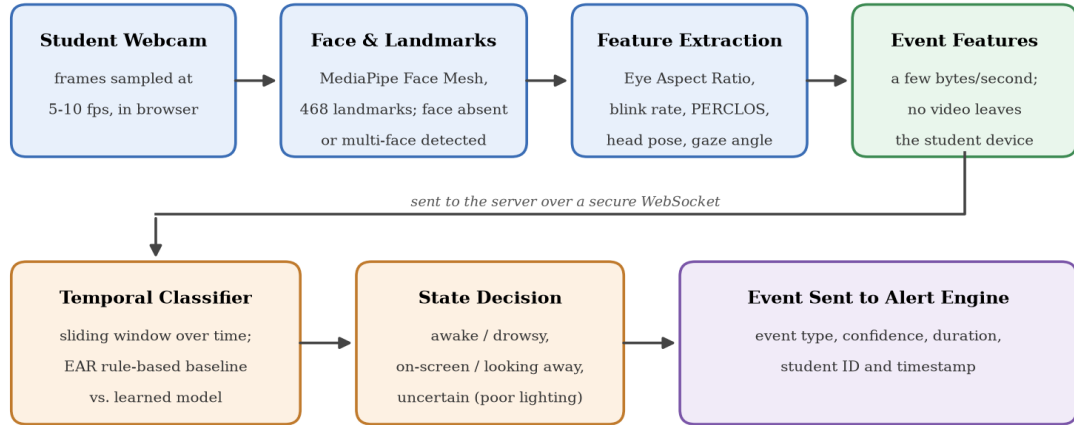


Figure 1: On-device vision pipeline for drowsiness and attention detection

7.3 Screen-Content Compliance and Slide Matching

The second objective compares what the student is viewing with what is being taught. The shared screen is sampled every few seconds, and a perceptual-hash filter discards frames that have not changed, which greatly reduces processing cost. Changed frames are passed through OCR (Du et al., 2020) [14] and image/text embeddings (Radford et al., 2021; Reimers & Gurevych, 2019) [15], [16], and the result is compared with the embedding of the instructor’s current slide and its neighbours. Because legitimate activities such as scrolling, loading a page or opening a reference produce brief mismatches, a low similarity score must persist beyond a configurable grace period before a compliance event is created.

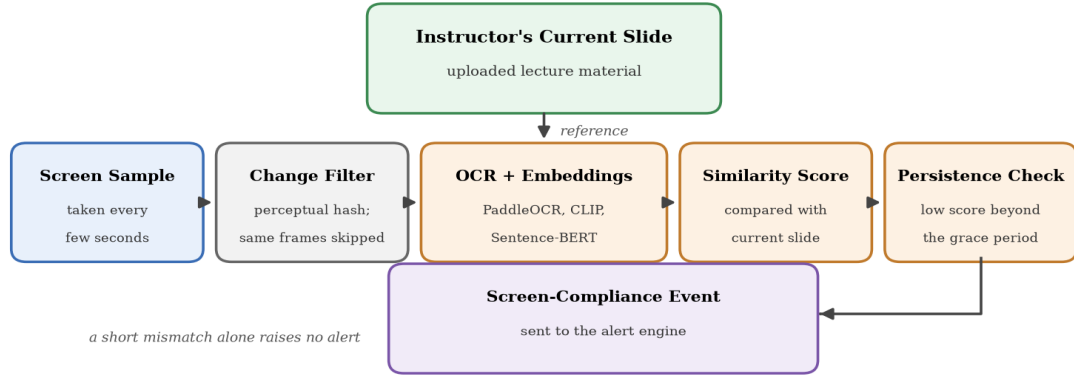


Figure 2: Screen-content compliance and slide-matching pipeline

7.4 Intelligent Alert Engine

The third objective is the component that turns raw detections into useful information. Every event carries the student identifier, event type, timestamp, confidence, duration and source module. An alert is produced only when the confidence exceeds a threshold, the condition persists longer than a configurable duration (for example drowsiness beyond five seconds, or looking away beyond twenty seconds), and the same alert is outside its cooldown window. Repeated events are grouped so that one struggling student cannot flood the dashboard, and alerts are ranked by severity. All thresholds will be tuned on a validation set and reported, rather than chosen by intuition.

The engine also supports a **private nudge**. When the instructor taps an alert, the system sends a short message and a gentle sound over the WebSocket channel of that single student, so the student is quietly brought back to the lecture while the rest of the class hears and sees nothing. The nudge is addressed to one session identifier instead of being broadcast to the room, it is rate-limited so that it cannot be misused, and every nudge is written to the audit log with its sender, receiver and time. The student sees who sent it and may reply with a one-tap response such as “I am back” or “I have a connection problem”, which is stored as feedback on that alert and later used to measure how often the alerts were justified.

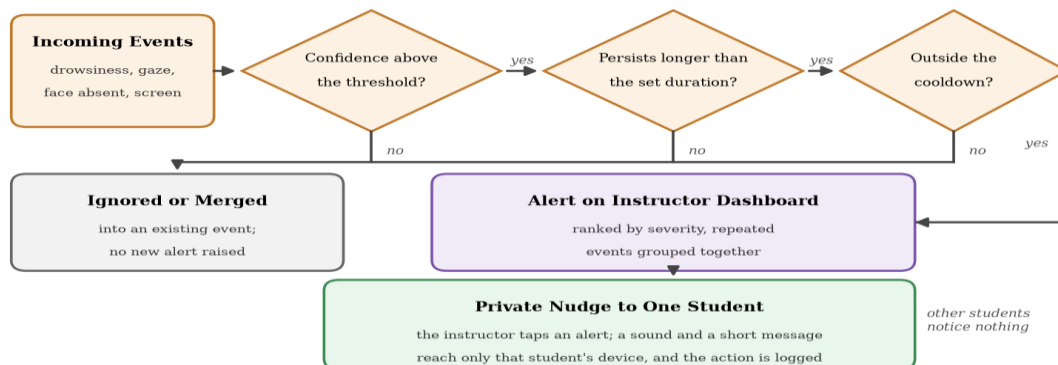


Figure 3: Alert engine decision flow and the private nudge to a single student

The dashboard is built entirely from these status signals. No video frame, photograph or screenshot of a student is ever transmitted to the instructor or stored on the server; the instructor sees a name, a roll number and a short state, as listed in Table 2.

Signal Shown	Meaning for the Instructor	Typical Trigger
Attentive	The student is present and following the lecture	Face detected and gaze towards the screen
Sleepy	The eyes have stayed closed for too long	EAR/PERCLOS beyond threshold for several seconds
Looking away	Attention has been off the lecture for a while	Gaze or head direction away beyond the set duration
No face	The student is not in front of the camera	No face detected for longer than the set duration
Camera off	The camera is switched off or blocked	No camera stream, or permission withdrawn
Wrong screen	The visible content no longer matches the slide	Low similarity beyond the grace period
Uncertain	Conditions are too poor to judge fairly	Weak landmark confidence, poor light or extreme angle
Connection problem	The student is losing the class, not ignoring it	Dropped or unstable real-time connection

Table 2: Status signals reported to the instructor (no video or images are sent)

7.5 Lecture Transcription and Learning Guidance

Lecture audio is transcribed in near real time with the open-source Whisper model (Radford et al., 2023) [17] running through faster-whisper. The transcript is segmented by topic, and a small open-source language model produces summaries, key points and likely examination questions. Slides and transcript chunks are embedded and stored in a vector database, so that a retrieval-augmented assistant (Lewis et al., 2020) [18] can answer questions using only that lecture’s material, citing the source slide and replying that a topic was not covered when the retrieved context is insufficient.

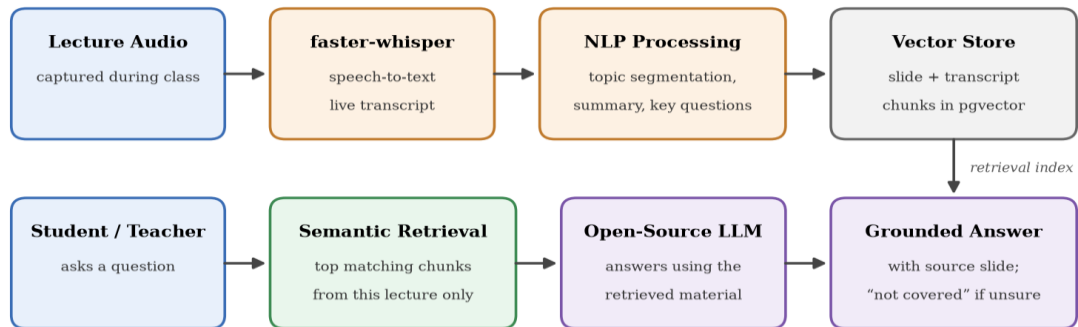


Figure 4: Lecture transcription, summarisation and retrieval-augmented assistance

7.6 System Architecture

Figure 5 shows how the components fit together. Student and instructor browsers communicate through the real-time layer and the application API; the AI services

produce events that pass through the alert engine to the instructor dashboard; and only structured data, never raw video, reaches the data layer.

IntelliClass AI – System Architecture

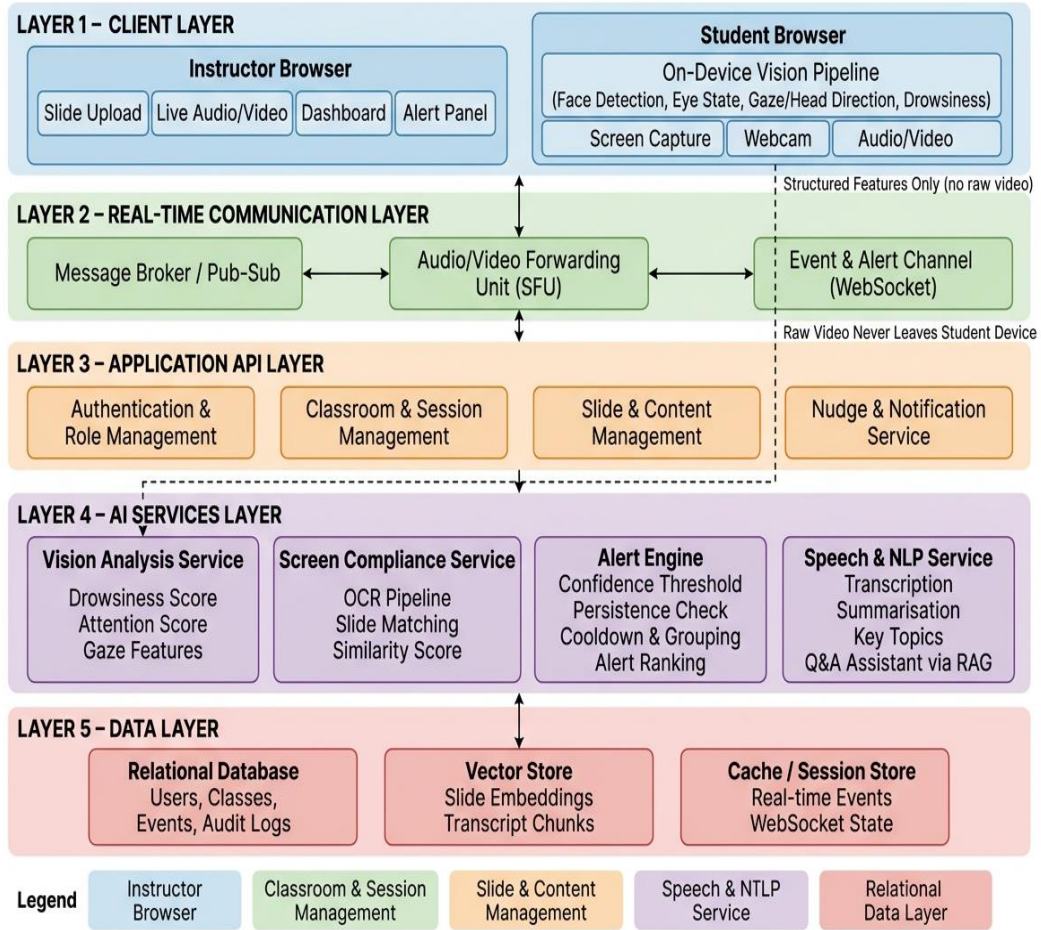


Figure 5: Block diagram of the proposed system

7.7 Evaluation Plan

Component	Evaluation Metrics
Drowsiness and attention models	Accuracy, precision, recall, F1-score, confusion matrix
Screen compliance	Precision, recall, F1-score, false-positive rate
Alert engine	Alert latency, false-alert rate, detection rate
Private nudge	Delivery success rate and latency, and confirmation that no other participant receives it
Speech-to-text	Word Error Rate (WER)
Lecture assistant	Answer correctness, groundedness, retrieval relevance
On-device vs. server inference	Latency, CPU and bandwidth usage per student
System and scalability	End-to-end latency and resource usage at 10, 25, 50 and 100 users

Table 3: Evaluation plan

7.8 Expected Research Contribution

- A real-time, multi-signal engagement pipeline that runs on the student device and is evaluated inside a live classroom rather than offline.
- An original screen-compliance dataset and evaluation protocol comparing student screen content with instructor slides, collected with consent.
- A confidence- and persistence-based alerting method, measured by its false-alert rate rather than by raw detection accuracy alone.
- An experimental comparison of client-side and server-side inference in terms of latency, bandwidth and scalability for online classrooms.

8. Tools and Technology

8.1 Software

All software used in this project is free and open-source, or is used within a free tier or free student plan. **No paid licence, paid API or paid hosting is required.**

Layer	Tools / Technologies (all free)
Frontend	Next.js, React, TypeScript, Tailwind CSS
Backend	Python, FastAPI, Pydantic
Real-time	WebRTC through LiveKit Cloud free tier (SFU with built-in TURN), WebSocket, Redis pub/sub
Computer Vision	MediaPipe (runs in the browser), OpenCV, PyTorch
OCR / Similarity	PaddleOCR, CLIP, Sentence-BERT (open-source models)
Speech-to-Text	faster-whisper (open-source Whisper, small/base model on CPU; no paid API)
LLM / RAG	Small open-source LLM (e.g. Llama 3.2 3B or Qwen2.5 3B) served on Hugging Face Spaces or locally through Ollama; Groq free tier for faster responses
Database	Supabase free tier (PostgreSQL with pgvector), Upstash free tier (Redis)
Frontend Hosting & Domain	Vercel free Hobby plan with a free <code>.vercel.app</code> subdomain (e.g. <code>intelliclass-ai.vercel.app</code>) and automatic HTTPS
Backend / AI Hosting	Hugging Face Spaces (free CPU, Docker) for the FastAPI backend and AI services; Supabase and Upstash hold all data because free Spaces storage is not permanent
DevOps	Docker, Docker Compose, Nginx, Let's Encrypt (free SSL), GitHub Actions
Testing	PyTest, Vitest, Playwright, k6/Locust (load testing)
Model Training	Google Colab and Kaggle Notebooks (free GPU)
Other	Git/GitHub (with GitHub Student Developer Pack), VS Code, Figma (free education plan)

Table 4: Software tools and technologies (zero-cost stack)

8.2 Hardware

- Team members' existing laptops/PCs: Intel Core i5/Ryzen 5 or above, minimum 8 GB RAM (16 GB recommended for running the local LLM).

- No dedicated GPU needs to be purchased; model training uses the free GPUs of Google Colab and Kaggle.
- Built-in laptop webcams and microphones for testing; team members' own devices for concurrent-user tests.
- Deployment on free infrastructure: Vercel (frontend), Hugging Face Spaces (backend and AI services), LiveKit Cloud (audio/video) and Supabase/Upstash (data), so there is no hosting cost.

Estimated project cost: Rs. 0. Every tool and service above is free or open-source, so the project needs no budget.

9. Mile Stone (Gantt Chart)

#	Milestone	Main Tasks	Duration
1	Requirement Analysis & Literature Review	Requirements, literature, datasets	5 weeks
2	System Design	Architecture, database schema, UI mock-ups	4 weeks
3	Environment Setup, Auth & Roles	Repo, Docker, database, login/RBAC	3 weeks
4	Virtual Classroom	WebRTC/SFU, WebSocket, dashboards	6 weeks
5	Vision Module	Face, eye, drowsiness, gaze; model training	9 weeks
6	Screen Compliance	Sampling, OCR, slide matching	6 weeks
7	Alert Engine & Dashboard	Thresholds, cooldown, alerts, private nudge	5 weeks
8	Speech, NLP & RAG Assistant	Whisper, summaries, questions, RAG	7 weeks
9	Integration & Deployment	Service integration, deployment on Vercel and Hugging Face Spaces	3 weeks
10	Testing & Evaluation	Model metrics, load tests, user testing	5 weeks
11	Documentation & Final Demo	Final report, user manual, demonstration	4 weeks

Table 5: Project milestones (several tasks overlap)

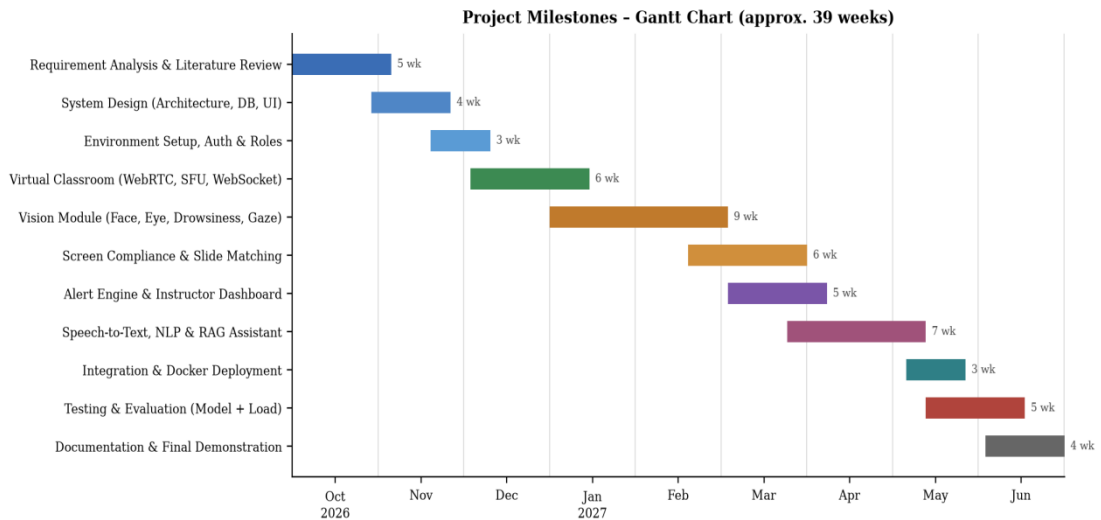


Figure 6: Gantt chart of the project timeline (October 2026 – June 2027)

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